

MGTA 495: AI & Prescriptive Analytics

WINTER 2026

CLASS TIME: Wednesday, 8:00 am – 10:50 am

LOCATION: OTRSN 1E107

PROFESSOR: Pooyan Kazemian

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OFFICE HOURS: Wednesday 11:15 am – 12:15 pm, or by appointment

TEACHING ASSISTANT: [TBD]

TA EMAIL: [TBD]

DESCRIPTION

This course provides an integrated introduction to Artificial Intelligence (AI), optimization, and simulation techniques that form the foundation of prescriptive analytics. In the first half of the course, students explore core AI and deep learning methods that enable data-driven prediction, pattern recognition, computer vision, natural language processing (NLP), and large language models (LLMs). We will also discuss model interpretability, fairness, and the mitigation of algorithmic bias in AI models. The second half of the course transitions to prescriptive analytics and discusses how simulation and optimization methods can guide data-driven analytics and decision-making in business. Topics include mixed-integer programming (MIP) for resource allocation and planning, dynamic programming for sequential decision-making under uncertainty, and discrete-event simulation for modeling complex systems in operations, finance, and healthcare. Through a combination of lectures, real-world examples, and hands-on coding labs, students will gain a practical and conceptual understanding of how modern AI, simulation, and mathematical modeling techniques can be applied to optimize processes, improve outcomes, and support strategic decision-making in various business domains.

OBJECTIVES

Upon completion of this course, students will be able to:

- Understand foundational concepts of artificial intelligence (AI), mathematical optimization, and simulation modeling, and how they can help guide business decisions.
- Apply deep learning methods, including multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and natural language processing models, to solve prediction and pattern recognition problems across various business domains.
- Evaluate the interpretability, fairness, and ethical implications of AI models, identifying potential sources of bias and employing strategies to improve transparency and accountability in AI-driven decision support systems.
- Build and analyze discrete-event simulation (DES) models to assess system performance under uncertainty and compare alternative strategies.
- Formulate and solve prescriptive analytics problems using mathematical optimization techniques, including linear, integer, and dynamic programming, to support data-driven decision-making.
- Critically assess analytical models and results, understanding the underlying assumptions, strengths, and limitations of AI, optimization, and simulation methods when applied to complex, real-world decision problems.

MATERIALS

The instructor will provide all required materials, including slides, datasets, and coding exercises, which will be accessible through Canvas. The following resources have been used in developing the course content and are recommended for optional additional reading:

- Chollet, F. (2018). Deep Learning with Python. Shelter Island, NY: Manning Publications
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. Cambridge, Massachusetts: The MIT Press.
- Agrawal, A., Gans, J., & Goldfarb, A. (2022). Prediction Machines, Updated and Expanded: The Simple Economics of Artificial Intelligence. Harvard Business Press.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Application in R. New York: Springer. Springer Science Business Media
- Hillier, F. S., & Lieberman, G. J. (2015). Introduction to operations research. McGraw-Hill.
- Banks, J. (2005). Discrete event system simulation. Pearson Education India.
- Law, A (2014). Simulation modeling and analysis. McGraw-Hill.
- SimPy User Manual (<https://simpy.readthedocs.io>)

SCHEDULE

Date	Class Topics	Due
Week 1	Introduction to Artificial Intelligence (AI) Different Types of AI; Deep Neural Networks; Multilayer Perceptron (MLP)	
Week 2	Improving the Performance of Neural Network Models Model Generalization and Overfitting; Regularization Methods	Assignment 1 [C]
Week 3	Computer Vision Convolutional Neural Network (CNN); Data Augmentation; Transfer Learning; Fine-Tuning	
Week 4	Natural Language Processing (NLP) Recurrent Neural Network (RNN); Generative AI and LLMs	Assignment 2 [C]
Week 5	Deep Learning for Sequential Data LSTMs, Transformers, Combining CNNs and RNNs	
Week 6	Explainable AI and Machine Bias Supplementary Topics in Deep Learning and AI	Assignment 3 [C]
Week 7	Discrete Event Simulation (DES) Building and Analyzing DES Models with SimPy; Decision Making with Simulation; Resource Allocation	Midterm Exam (Take-Home) [I]
Week 8	Input and Output Data Analysis for Simulation Modeling Distribution Fitting; Comparing Alternative Scenarios	
Week 9	Mixed-Integer Programming (MIP) Branch-and-Bound Algorithm; Formulating and Solving MIP Problems with DOcplex; Examples and Applications of MIP	Assignment 4 [C]
Week 10	Dynamic Programming States, Actions, Objective Function; The Principle of Optimality; Solving a Dynamic Program	Assignment 5 [C]
Week 11	Final Exam	Final Exam [I]

ASSIGNMENTS

All assignments will be posted on Canvas with a clear due date. Key:

I – Independent, individual work only. No collaboration or consultation allowed.

G – Students may work together in groups and turn in one project or assignment for the entire group.

C – Collaboration with classmates is allowed. However, each student must submit]

GRADING

Exams and Deliverables	Percentage
Homework Assignments	30%
Class Attendance & Participation	20%
Midterm Exam	20%
Final Exam	30%
Total	100

COURSE POLICIES

Course Format. The course will be offered in person and will consist of 10 weekly sessions. Sessions will include interactive lectures and hands-on coding exercises in Python. Students are encouraged to bring their laptops to follow along with lecture notes and coding exercises. All lectures will be recorded.

Attendance Policy. In-person attendance is mandatory for all students. If you must miss a class due to a valid reason (e.g., illness or family emergency), you must email the instructor with appropriate documentation (proof of valid reason) and watch the lecture recording in full within 7 days of the missed class to receive the attendance credit. Missing more than 3 classes will result in a failing grade (“F”) for the course – no exceptions.

Expectations and Late Submissions. Students will be expected to complete five homework assignments, participate in class discussions, and take a midterm exam (take-home) and a final exam. All assignments must be turned in via Canvas by the specified deadline. Late submission will incur a penalty of 10% of the total points for each 24-hour period (or part thereof) past the deadline, and submissions are not accepted after solutions are posted on Canvas (usually 2-3 days after the due date). Students are encouraged to submit their assignments early if they anticipate delays closer to the due date.

Use of AI Tools such as ChatGPT. Generative AI (e.g., ChatGPT) can be used to brainstorm ideas for your assignments and assist in debugging your code. However, you should understand and explain every step of your analysis in your submission. You should also fully understand and be able to explain all details of every line of code that you submit as your own work. Any other use of GenAI requires preapproval and must be documented and submitted along with the relevant assignment. Failure to submit documentation or identify use of AI is considered an integrity violation:
<https://academicintegrity.ucsd.edu/excel-integrity/gen-ai/index.html>

Changes. The instructors reserve the right to make changes during the term to any aspect of the syllabus that, in their judgment, are needed to achieve the learning objectives of the course.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.